

TechMentor AI: A Multi-Agent Intelligent Learning and Interview Assistant for Computer Science Education

Project Synopsis

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Introduction

Computer Science students pursuing graduate and undergraduate programs often struggle to find interactive, subject-specific, and context-aware academic assistance. Static textbooks, recorded lectures, and general-purpose chatbots fail to address individualized doubts, assess student understanding in real time, or simulate the technical interview environment that students need to prepare for placements and competitive examinations such as GATE.

TechMentor AI addresses this gap by proposing a web-based, multi-agent intelligent platform that enables students to resolve doubts across core Computer Science subjects, take auto-generated quizzes, practice mock technical interviews, and receive personalized study roadmaps — all through a single, conversational interface. The system combines Transformer-based NLP models, Sentence-BERT semantic embeddings, and a coordinated multi-agent architecture built with CrewAI and Streamlit to deliver a responsive and educationally meaningful experience.

Objectives

The primary objectives of the project are:

- To design and implement a multi-agent architecture where a Supervisor Agent routes user queries to specialized subject and task agents.
- To develop subject-specific agents for Operating Systems, DBMS, Computer Networks, and Data Structures and Algorithms that answer student queries accurately.
- To build a Quiz Generation Agent that dynamically creates MCQs and practice questions for knowledge assessment.
- To create an Interview Preparation Agent that conducts mock technical interviews, generates questions, and provides feedback on student answers.
- To implement a Study Roadmap Agent that generates personalized, topic-wise study plans based on student interaction history.
- To deploy the complete system as a publicly accessible web application on Hugging Face Spaces.

Subjects and Domains Supported

The platform currently supports the following Computer Science domains:

- Operating Systems – scheduling, deadlocks, memory management, synchronization
- Database Management Systems (DBMS) – normalization, indexing, transactions, SQL
- Computer Networks – OSI/TCP-IP models, routing, protocols, congestion control

- Data Structures and Algorithms – arrays, trees, graphs, sorting, dynamic programming
- Technical Interview Preparation – mock interviews, HR and coding questions, feedback
- GATE Exam Preparation – topic-focused practice and revision

Proposed System Architecture

The system follows a modular, agent-based architecture optimized for educational task delegation and rapid web deployment:

1. Streamlit Frontend

A browser-based interactive interface built entirely in Python using Streamlit. Students enter queries, select subjects, attempt quizzes, and view study roadmaps through this single-page interface, eliminating the need for a separate frontend framework.

2. Supervisor Agent

The central orchestrator of the system. It receives all user inputs, performs NLU-based intent classification, and delegates requests to the appropriate specialized agent. It also manages session context and ensures conversational continuity across multi-turn interactions.

3. Subject-Specific Agents

Dedicated agents for OS, DBMS, Computer Networks, and DSA, each equipped with domain-specific knowledge prompts and retrieval-augmented generation capabilities. They provide accurate, context-aware responses to subject queries using Transformer-based language models.

4. Quiz Generation Agent

Generates multiple-choice questions and short-answer assessments dynamically based on specified topics or the student's recent interaction history, enabling active recall and self-assessment.

5. Interview Preparation Agent

Simulates technical interview sessions by generating domain-specific interview questions, evaluating student responses, and providing structured feedback on correctness, completeness, and clarity.

6. Study Roadmap Agent

Creates a personalized, time-bound study plan based on student interactions and assessed performance. It prioritizes weak areas and recommends specific topics and resources for improvement.

7. Session Storage (SQLite)

SQLite is used to store user session data, quiz results, and interaction history, enabling progress tracking and informing roadmap generation without requiring a heavy database backend.

Technologies Used

Component	Technology
Frontend / UI	Streamlit (Python)

Agent Orchestration	CrewAI
Language Models	Hugging Face Transformers
Semantic Embeddings	Sentence Transformers (all-MiniLM-L6-v2)
NLP / NLU	Transformer-based models, intent classification
Database	SQLite
Deployment	Hugging Face Spaces
Language	Python

Working Methodology

The system workflow proceeds as follows:

- Student enters a natural language query or selects a task (doubt solving, quiz, interview, roadmap) via the Streamlit interface.
- Supervisor Agent receives the input, classifies intent using NLU, and routes the request to the appropriate specialized agent.
- The relevant agent processes the query using Transformer-based language generation and Sentence-BERT semantic retrieval.
- For quizzes, the Quiz Generation Agent produces MCQs; for interviews, the Interview Agent generates questions and evaluates answers.
- The Study Roadmap Agent reads session history from SQLite and generates a personalized, topic-prioritized study plan.
- All responses are displayed to the student in the Streamlit interface in real time.

Expected Outcomes

The proposed system is expected to:

- Provide accurate, context-aware answers to student queries across five core Computer Science subjects.
- Generate dynamic quizzes and instantly evaluate student understanding with feedback.
- Simulate realistic technical interview sessions and provide actionable improvement feedback.
- Produce personalized study roadmaps that help students prioritize learning efficiently.
- Demonstrate practical integration of Multi-Agent Systems, NLP, and Transformer models in an educational AI application.

Future Enhancements

- Voice-based interaction using speech-to-text and text-to-speech modules for a more natural experience.
- Expansion of knowledge base to cover Machine Learning, Software Engineering, and Theory of Computation.
- Reinforcement learning for adaptive study roadmap optimization based on longitudinal performance data.
- Multilingual support to make the platform accessible to a broader student population.
- Integration with Learning Management Systems (LMS) such as Moodle for institutional adoption.

- Upgrade from prompt-based agents to full LangChain agents with memory and tool-use capabilities.

Conclusion

TechMentor AI combines a multi-agent architecture, Transformer-based NLP, Sentence-BERT semantic embeddings, and modern AI orchestration frameworks to create an intelligent, transparent, and practical educational assistant for Computer Science students. By using Streamlit and CrewAI on a deployable Hugging Face Spaces infrastructure, the system is designed to be fully implementable within a single academic semester without sacrificing core functionality.

This project demonstrates the practical application of AI agents in education technology and showcases the integration of Natural Language Processing, Multi-Agent Systems, and generative AI in a domain-specific, student-centric platform.
